Receiver-Conditioned Latent Uncertainty Propagation for Multi-Agent Web Reasoning

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Abstract

Role-specialized language-model agents retrieve, interpret, and combine Web evidence through dependent intermediate decisions. These dependencies can transmit an erroneous premise together with misleading confidence, while a sender's uncertainty may concern a different prediction event from the receiver's. Calibrating local scores alone therefore neither determines downstream reliability nor identifies what should be clarified. We present Receiver-Conditioned Latent Uncertainty Propagation (RCLUP), which couples candidate-specific latent messages with task-conditioned risk estimation and targeted inter-agent clarification. A dependency-aware receiver jointly reads alternative interpretations, their source lineage, and the receiving task, rather than combining incoming confidence scores as independent evidence. Action-conditioned predictors estimate a common, policy-specified final-error event, and held-out role-conditional calibration gives those estimates comparable probabilistic semantics. The receiver uses the predicted benefit of a concrete clarification to decide whether to query a sender, then recomputes its decision from the returned evidence. The evaluation protocol separates inherited-error propagation, calibration, and clarification utility, with matched text, generic-latent, and intervention-aware controls. Numerical results remain unreported in this pre-experimental draft.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Question answering**.

Keywords

multi-agent systems, uncertainty propagation, calibration, latent communication, clarification, Web question answering

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1 Introduction

Web information seeking increasingly involves more than retrieving a relevant page: a system must identify the intended entity, reconcile evidence across sources, and derive an answer whose supporting assumptions remain valid. Role-specialized large language model (LLM) agents offer a natural division of this work. A retrieval agent selects documents, an interpretation agent resolves entities and extracts claims, and a reasoning agent combines those claims before a verifier checks the result. Their outputs are not independent answers to the same question. They are different intermediate artifacts that become premises for subsequent tasks. Consequently, a mistaken interpretation can travel beyond its point of origin and affect an otherwise competent downstream agent. Reliability in agentic Web question answering therefore depends on what is preserved, questioned, and revised at these interfaces, not only on the quality of each isolated agent.

These dependencies create two connected uncertainty problems. First, an erroneous premise can be forwarded together with unwarranted confidence, and several agents may appear to corroborate it while relying on the same upstream evidence. Propagation-aware risk estimation and communication-dependent calibration have begun to address these effects [5, 7]. Second, the uncertainty being transmitted may not describe the event that the receiver needs to predict. A document selector's uncertainty over entity identity and a reasoning agent's uncertainty over a temporal comparison refer to different tasks. Even perfectly calibrated local error probabilities need not imply equal downstream consequences. If two candidate dates lie on opposite sides of a threshold, resolving their difference matters to that comparison; if both imply the same result, the same upstream uncertainty may be immaterial. Reliable propagation must therefore account for both communication dependencies and the receiver's actual task, rather than rescaling a sender's confidence and treating it as downstream reliability.

Identifying downstream risk still leaves an action problem. Two equally risky handoffs may call for different clarifications: one requires distinguishing homonymous entities, whereas another requires distinguishing an incorporation date from the start of operations. A risk score does not specify which of these questions is useful. More generally, predicting failure and predicting the benefit of an intervention are distinct objectives [11]. An agent that asks whenever confidence is low may waste communication on irrelevant alternatives, while an agent that trusts a confidently stated premise may omit the question needed to correct it. Effective uncertainty propagation must preserve the distinctions that explain how an inherited judgment could change the receiver's decision and make those distinctions available for selecting a concrete clarification.

Latent communication offers one way to retain information beyond a selected textual artifact. CIPHER communicates probability-weighted embeddings; LatentMAS transfers latent thoughts through shared working memory; and Interlat learns receiver-side adaptation and task-oriented latent compression [2, 8, 13]. Interlat also incorporates uncertainty-aware behavior preservation, so task relevance and uncertainty are not absent from existing latent communication. What remains to be established for role-specialized workflows is whether candidate-associated latent content supports *calibrated, receiver-specific clarification* beyond generic latent transfer and sufficiently informative text. Hidden states are not evidence certificates, and richer communication alone does not guarantee that an agent identifies the right uncertainty or uses its resolution

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correctly. This distinction motivates studying how inherited representations are organized and interpreted, rather than assuming that transmitting more latent information solves propagation.

We propose *Receiver-Conditioned Latent Uncertainty Propagation* (RCLUP) to connect these requirements. A sender associates its alternative artifacts with compact latent representations and explicit provenance. The receiver jointly processes these alternatives using its own task and the communication graph, retaining shared-source and shared-ancestor relationships rather than treating incoming confidence as independent support. It then estimates the risk of a specified final-error event under continuing or making a particular clarification. Role-conditional calibration aligns the meaning of these predictions without conflating local correctness, downstream error, and intervention value. A clarification targets a particular sender and candidate distinction; returned evidence produces a new version of the handoff and a fresh receiver decision. Thus, latent propagation supplies decision-relevant content, calibration supplies a defined probabilistic interpretation, and clarification provides an action through which inherited uncertainty can be resolved.

The framework contributes a candidate-indexed communication mechanism that retains task-relevant inherited content across multiple handoffs, a dependency- and role-conditioned treatment of downstream reliability, and a clarification policy tied to concrete candidate distinctions. We evaluate these contributions separately: whether erroneous premises persist less often under dependency-aware reception, whether a common final-error target is better calibrated across roles, and whether candidate-specific latent content improves clarification beyond rich text and matched generic latent communication. Fixed-prefix comparisons isolate risk estimation and question choice; closed-loop evaluations determine whether these changes improve final answers under comparable interaction budgets. The numerical results are left open until these tests are completed.

2 Related Work

Uncertainty under agent dependencies. PropUQ-MAS combines locally estimated risk with risk inherited through communication edges, including the receiver's tendency to accept upstream information [7]. CAGE-Cal models the difference between communicating and counterfactual non-communicating agent graphs to calibrate panel confidence [5]. These works motivate retaining dependency information, but their risk targets do not by themselves specify which alternative interpretation should be clarified in a receiver's different subtask. RCLUP uses a common terminal-error event for cross-role risk comparison and retains candidate content for action selection. It neither claims that previous propagation ignores receiver behavior nor identifies an observational risk score with the causal effect of a particular message.

Latent representations and communication. Hidden-activation probes predict statement truthfulness, and semantic entropy probes predict semantic uncertainty from a single generation [1, 6]. These establish distinct uses of internal reliability signals, not a guarantee that every hidden feature is uncertainty-relevant. CIPHER, LatentMAS, and Interlat investigate continuous inter-agent communication [2, 8, 13]. We distinguish candidate-indexed uncertainty content from generic latent capacity by giving both representations the same receiver and clarification supervision. Explicit candidate distributions, source spans, and uncertainty explanations are also necessary controls; a comparison against a bare confidence score cannot establish latent-specific value.

Calibration and uncertainty-guided intervention. Post-hoc calibration can improve the interpretation of predictive scores [4], while AUQ uses confidence and explanations to guide reflection [12]. Work on intervention value shows why a failure predictor alone need not select useful actions [11]. Accordingly, we predict the outcome after each concrete action instead of using risk as an intervention threshold. The claimed distinction is not the use of intervention supervision itself, but whether receiver-conditioned candidate latents provide additional information for selecting a source-grounded clarification under the same supervision and budget.

3 Problem Formulation

3.1 Role-Specialized Handoffs

An execution is a time-unrolled directed acyclic graph $G_t = (V_t, E_t)$. A node is an *invocation*, not a persistent agent identity. Invocation i has role r_i, subtask T_i, accessible evidence D_i, and history H_i; it produces a local artifact $a_i \in \mathcal{A}_i$. The spaces $\mathcal{A}_i$ differ: document sets, grounded fact tuples, derived answers, and verification decisions are not interchangeable labels. An edge $i \to j$ means that receiver j uses i's artifact. A backward clarification creates a later invocation and a new message version, so the unrolled graph remains acyclic. All agents retain their authorized tools, history, checks, and ordinary repair actions.

Let S_j denote the public decision state at a handoff, including T_j, the receiver's evidence, its incoming messages, remaining budgets, and provenance. A sender's self-reported probability is an observation in S_j, not trusted ground truth. If local reference labels exist, its local error is $E_i^{\text{local}} = \mathbb{I}[a_i \text{ violates its local task}]$. This event is different from a final workflow error. We do not compare entropy values across different output spaces or interpret normalized sample frequencies as calibrated correctness.

3.2 A Common Risk Target, Not a Common Entropy Scale

We fix a reference continuation policy π_0: it completes the remaining role-specialized workflow using the shared backbone, tools, verification, and standard repair, without further *learned handoff clarifications*. Disabling this particular research controller does not disable ordinary recovery. For an action u at S_j, let $E_{j,u}^{\pi_0} = 1$ when the eventual answer is incorrect according to the frozen benchmark validator, and 0 otherwise. Missing or unresolved final answers count as errors in this endpoint. The action-conditioned risk is

$$r_j(u) = \Pr(E_{j,u}^{\pi_0} = 1 \mid S_j, u). \tag{1}$$

Thus, risks at different roles concern the *same final-error event*, conditioned on different information and actions. They do not assign the same importance to equally uncertain local predictions. They are also policy-relative forecasts, not causal blame assigned to an upstream agent.

The action set contains commit(b) for a receiver artifact b, and ask(q) for a particular clarification. The latter includes obtaining the answer, updating the handoff, selecting a receiver artifact with the frozen receiver module, and then following π_0. We define a separate final-output risk $r^{\text{final}} = \Pr(E^{\text{final}} = 1 \mid S^{\text{final}}, \hat{y})$ for the answer actually produced by the deployed closed-loop policy. Equation (1) is never reused as if it were this post-intervention probability. This separation prevents evaluating a revised answer with an unrevised risk score.

Two failure boundaries. **Dependency-induced misconfidence** arises when inherited errors or repeated evidence are treated as fresh support. **Task mismatch** occurs when local sender risk substitutes for a probability about the receiver's outcome. Correcting the second requires task-conditioned prediction and calibration; correcting the first additionally requires preserving which evidence and alternatives the receiver actually depends on. Neither correction alone identifies the best clarification, which requires action-specific outcomes.

4 Method

4.1 Candidate-Specific Latent Messages

At invocation i, a frozen backbone generates S local artifact proposals using only (r_i, T_i, H_i, D_i). A role-specific canonicalizer groups equivalent serialized artifacts. We retain the greedy proposal and the $K-1$ most frequent distinct alternatives; discarded proposals form a residual category $\perp$. For retained candidate k, n_{ik} is its sample count and

$$w_{ik} = \frac{n_{ik} + \alpha}{S + \alpha(K_i + 1)}, \qquad w_{i\perp} = \frac{S - \sum_k n_{ik} + \alpha}{S + \alpha(K_i + 1)}. \tag{2}$$

Here $K_i \le K$ is the number retained. These weights describe model support over proposed artifacts, not the probability that every possible correct answer is represented. Keeping the residual category exposes, rather than hides, candidate truncation.

For each retained candidate, the representative generation with the highest length-normalized log probability supplies last-layer response states $H_{ik} \in \mathbb{R}^{L_{ik} \times d_B}$. These states may encode evidence processed during that generation but are not themselves source citations. A learned attention pooler compresses them into m vectors:

$$Z^{\text{loc}}_{ik} = \text{softmax}\left(\frac{Q(H_{ik}W_K)^\top}{\sqrt{d}}\right) H_{ik}W_V \in \mathbb{R}^{m \times d}. \tag{3}$$

The queries $Q \in \mathbb{R}^{m \times d}$ and projections are trained with the receiver; the backbone remains frozen. The residual candidate has a learned null representation, a missing-content flag, and no fabricated source span. A root invocation sends $Z_{ik} = Z^{\text{loc}}_{ik}$. Later invocations also carry forward inherited content using Equation (6).

A message is $M_i = \{(a_{ik}, w_{ik}, Z_{ik}, \mathcal{P}_{ik})\}_k$, together with the selected artifact and any reported risk's event and policy identifier. $\mathcal{P}_{ik}$ contains origin invocation, message version, cited document/span identifiers, and ancestor invocation identifiers. The serialized candidate and citations are available to every matched baseline. Only the candidate-associated hidden content differs in the representation comparison. In particular, we transmit neither one uncertainty scalar per candidate nor unqualified claims that the latent contains verified evidence.

4.2 Dependency-Aware, Receiver-Conditioned Reading

The receiver first encodes its own role, actual subtask, evidence, and incoming readable artifacts into $s_j \in \mathbb{R}^d$ using the frozen backbone and a trainable projection. The frozen decoder proposes receiver artifacts $\mathcal{B}_j$ in the receiver's output space. A schema-valid UNRESOLVED artifact is included so that candidate omission does not force an unsupported commitment; unresolved termination still counts as failure. A backbone proposal may request the same normal search or repair available to the controls; its cost is not suppressed.

Incoming latent packets with exactly the same origin and version are coalesced. Distinct messages are *not* merged merely because their answers agree. Let x_u be one candidate latent slot augmented with the candidate-text encoding, sender-role encoding, slot index, and residual flag. We jointly contextualize all slots with attention whose logits include dependency features:

$$A_{uv} = \frac{(x_u W_Q)(x_v W_K)^\top}{\sqrt{d}} + \eta^\top d_{uv}, \tag{4}$$

where d_{uv} records same origin, direct ancestry, ancestor-set overlap, and cited-source overlap. Two residual attention/MLP blocks yield contextual slots h_u. Source overlap and ancestor overlap are features of dependence, not proof that two claims are identical. Unknown provenance has its own mask. No independence product or multiplication of incoming correctness probabilities is used.

Each receiver artifact b queries this joint representation in the context of s_j:

$$\begin{aligned} q_{jb} &= W_q[s_j; \text{enc}(b)], \\ \beta_{jbu} &= \text{softmax}_u\left(q_{jb}^\top W_k h_u / \sqrt{d} + \log w_u\right), \\ c_{jb} &= \sum_u \beta_{jbu} W_v h_u. \end{aligned} \tag{5}$$

Here w_u is the support weight of the candidate owning slot u. The readout $\xi_{jb} = [s_j; \text{enc}(b); c_{jb}; v_j]$ also includes public graph summaries v_j: incoming edge count, unique-origin count, maximum source overlap, depth, and candidate-support entropy. The commit predictor produces a logit $\ell_j(b) = f_\theta(\xi_{jb})$ for the terminal error in Equation (1). Candidate commit risks are not normalized into an answer posterior: several different artifacts can all support a correct continuation. The reader does not map a sender label directly onto a receiver label. Rather, it evaluates receiver artifacts using the alternatives and evidence relevant to that receiver's task.

Propagating beyond one handoff. For each outgoing artifact b, the receiver combines its local candidate states with the inherited task-conditioned context:

$$\begin{aligned} \gamma_{jb} &= \text{sigmoid}\left(W_g[s_j; \text{enc}(b); c_{jb}] + b_\gamma\right), \\ Z^{\text{out}}_{jb,t} &= \text{LN}\left(Z^{\text{loc}}_{jb,t} + \gamma_{jb} \odot W_o c_{jb}\right), \quad t = 1, \dots, m. \end{aligned} \tag{6}$$

The next edge carries $Z_{jb} = Z^{\text{out}}_{jb}$ and the union of consumed origin/version identifiers. Root invocations use zero inherited context. The gate and projection are trained with the receiver; all consumed parents remain in provenance regardless of attention weight. This recursive carrier prevents inherited content from disappearing when the receiver becomes a sender. The learned readout selects

among frozen-decoder proposals and forms outgoing messages; it does not require injecting latent tokens into the backbone decoder. Thus the mechanism is distinct from an assumption that all reasoning occurs in latent space.

This construction distinguishes exact message duplication from shared-source dependence. Exact duplication is removed by identity. Residual correlations among distinct, re-encoded messages are learned from joint features and outcome supervision; we do not claim an exact Bayesian correction or immunity to correlated hallucinations.

4.3 Role-Conditional Calibration

A score trained across roles can still be miscalibrated within a particular handoff type. We therefore fit a partially pooled logistic calibrator on disjoint calibration questions:

$$\widehat{r}_j(u) = \mathrm{sigmoid}\big(e^{a_0+a_g}\ell_j(u) + b_0 + b_g\big), \quad g = (r_j, \mathrm{family}(u)). \tag{7}$$

The family distinguishes commit and clarification actions. Parameters minimize held-out Bernoulli negative log likelihood with an L_2 penalty on group deviations (a_g, b_g). Sparse or unseen groups use the global parameters. The positive slope preserves ordering within a group, while role-conditioned offsets and slopes adjust probability semantics. Consequently, calibration itself cannot explain improved within-role candidate ranking. Learned receiver content and calibration are evaluated separately.

For commit actions, $\ell_j(u) = \ell_j(b)$; for questions it is produced by the clarification predictor below. The receiver's provisional commitment is $b_j^* = \arg\min_{b\in\mathcal{B}_j} \widehat{r}_j(\mathrm{commit}(b))$. The calibrator targets the common final-error event, rather than forcing entropy or heterogeneous local error rates to coincide. It is an empirical, distribution-dependent calibration procedure, not a guarantee of group calibration for unseen roles or adaptively selected actions. We audit reliability on selected actions separately.

4.4 Clarifying the Relevant Candidate Distinction

A shared frozen proposer generates at most $Q_{\max}$ atomic questions from the receiving task, readable candidate alternatives, source identifiers, and existing question history. A question q specifies a sender i, a candidate pair (k, k'), its text, and the evidence scope. A schema checker rejects compound or unaddressable questions. It does not inspect answer labels. The matched controls receive the same menu; this isolates question *selection* from question generation. If the useful question is missing, no selector can recover it, so menu coverage is an explicit diagnostic.

The question predictor uses s_j, the question encoding, joint packet summaries, the two candidate means $\bar{h}_{ik}, \bar{h}_{ik'}$, their difference and elementwise absolute difference, their support weights, and their receiver-attention masses. A two-layer MLP g_κ yields $\ell_j(q)$. Thus, the same sender ambiguity can be scored differently for different tasks, and equal scalar risks need not imply the same question. After calibration, the estimated net gain is

$$\widehat{G}_j(q) = \widehat{r}_j(\mathrm{commit}(b_j^*)) - \widehat{r}_j(\mathrm{ask}(q)) - \lambda c(q). \tag{8}$$

The main protocol assigns $c(q) = 1$ per atomic information item and enforces a separate inference budget. This expected-value comparison is a control objective, not a new definition of uncertainty [11]. The system asks $\arg\max_q \widehat{G}_j(q)$ only if the gain is positive and both budgets permit it; otherwise it commits b_j^*. Ties favor fewer questions and then deterministic candidate order.

The sender responds using its authorized evidence and tools. The response includes a direct answer, document/span identifiers, and an unresolved flag. A provenance check verifies source membership and quote offsets, not semantic truth; the common verifier and normal repair remain available. A failed or unresolved clarification is retained as such, not converted into support. New evidence triggers fresh sender proposals and carrier construction, replaces that sender's previous version, invalidates downstream decisions that consumed it, and reruns the receiver. Identical evidence-event identifiers are idempotent. A repeated unresolved question is not reissued against an unchanged source version. This is an evidence-conditioned recomputation, not an additive confidence bonus for having asked.

4.5 Training and Closed-Loop Inference

Training uses paired continuations from saved training prefixes, not invented question-value labels. For each receiver artifact b, we commit b and run the same π_0 under R preassigned seeds. Let e_{jbr} be the resulting binary final error; suffix generations use the fixed stochastic decoding policy in Appendix C. The pooler, dependency reader, and commit head are trained with

$$\mathcal{L}_{\mathrm{commit}}(\theta) = \sum_{j,b,r} \omega_j \ \mathrm{BCE}\left(\mathrm{sigmoid}\,\ell_j(b), e_{jbr}\right). \tag{9}$$

Weights ω_j give each original question equal total weight. All candidates are evaluated, not only the one eventually selected. Multiple locally valid artifacts may have similar terminal success; no artificial unique local answer is required. The decoder, tools, and reference continuation stay fixed. Saved backbone states are reused, while the trainable carrier is recomputed in ancestry order using only prefix information. Training also includes prefixes obtained after questions proposed independently of their outcomes, so recomputation states are represented.

We then freeze the receiver and construct clarification targets. For every admitted question, execute the real sender response, apply the same update rule, select the new artifact with the frozen receiver, and run π_0. With outcome e_{jqr},

$$\mathcal{L}_{\mathrm{ask}}(\kappa) = \sum_{j,q,r} \omega_j \ \mathrm{BCE}\left(\mathrm{sigmoid}\,\ell_j(q), e_{jqr}\right). \tag{10}$$

All proposal, response, and continuation work is charged. Seeds pair continuations but do not make their realized outcomes noise-free causal effects. These labels estimate action values inside the specified environment, not real-user clarification utility. Calibration then uses independent prefixes from the same reference-continuation protocol. Policy cost λ and budget settings are chosen on selection data, never on test outcomes.

At deployment, the controller reapplies Equation (8) after each new handoff. This is receding-horizon control: its calibrated risks concern one specified action followed by π_0, while the deployed

Algorithm 1 Receiver-conditioned propagation and clarification

Require: Role graph; frozen backbone/tools; trained modules; budgets

for each invocation j in execution order **do**
  Receive versioned candidate packets and readable artifacts
  repeat
    Coalesce identical packets; encode task and dependencies
    Generate receiver candidates; score/calibrate commit risks
    Set b_j^* to the minimum-risk admissible candidate
    Propose the common question menu; score/calibrate questions
    if a positive-gain question fits the remaining budgets **then**
      Ask its sender; record evidence, failures, and full cost
      Version the revised packet and invalidate affected states
    else
      Submit b_j^* through the common verifier and repair loop
      break
    end if
  until budget exhausted
  Emit the checked artifact and composed carrier, or unresolved output
end for
Score the actual final answer using the terminal calibrator

policy may choose later clarifications. We therefore train a separate terminal-error head on completed training trajectories of the frozen deployed policy and calibrate it on a second, independent calibration split. It reads the final post-update state and actual answer; its score, not an earlier continuation risk, accompanies the final answer. Calibration metrics for fixed-prefix actions and final outputs are reported separately.

5 Experimental Protocol

Status. This section specifies experiments to be executed. Every — in a result table means *not measured*, not zero; no improvement or significance is asserted.

5.1 Tasks, Roles, and Evidence Access

The primary tasks are HotpotQA and MuSiQue, which provide multi-hop questions and evidence annotations [9, 10]. DROP supplies a complementary discrete-reasoning setting with different extraction and calculation tasks [3]. HotpotQA and MuSiQue are the Web evidence-navigation experiments; DROP is a controlled reasoning-transfer evaluation, not independent proof of Web deployment. The main studies use released document snapshots and reproducible retrieval rather than a changing live search engine. Gold answers and support annotations are used only by offline validators and never as agent context, queries, or source-selection oracles.

Table 1: Roles differ in inputs, operations, and output spaces.

Role	Assigned subtask	Artifact passed onward
Retriever	Search for evidence relevant to a subquestion	Ranked source IDs and passages
Interpreter	Resolve an entity, date, definition, or relation	Grounded candidate fact tuples
Reasoner	Combine facts or execute a discrete operation	Derived answer and dependencies
Verifier	Check claim–source and computation consistency	Verdict, corrections, or unresolved issues

Table 1 defines functional specialization. A fixed decomposition prompt routes real subtasks, not different personas solving the same question. The protocol includes chains and branch–merge graphs in which parallel retrieval/interpretation paths may reuse a source. Graph construction follows the task decomposition rather than injecting incorrect messages. All methods have the same source index, fetching and search interfaces, calculator, complete necessary histories, verifier, and repair opportunities. Receivers can request a cited source directly; communication is not made useful by forbidding ordinary rereading.

The planned backbones are Qwen3-8B and Mistral-7B-Instruct-v0.3. Within an experiment, all roles use the same frozen backbone; task heterogeneity is not confounded with hidden-space alignment between model families. Each backbone receives separately trained small modules. Training/selection/calibration/test partitions are disjoint by original question and available composition family; all candidates, interventions, seeds, and paraphrases from a question remain together. A source-disjoint sensitivity analysis removes training questions sharing annotated support documents with evaluation questions. It does not remove hard test questions after seeing outcomes. Exact counts and source hashes will accompany executed runs.

5.2 Comparators and Matched Controls

The main comparison separates representation, dependence, calibration, and control. *Text MAS* uses the normal complete workflow. *Calibrated scalar* adds local confidence and task/role-conditioned scalar risk. *Rich text* transmits the same candidates, weights, provenance, source spans, and structured uncertainty explanations. *Generic latent* uses the same pooling budget and trainable receiver, but pools all candidate traces jointly without candidate association. Its optimizer, labels, and parameter budget match RCLUP. All learned variants receive the same branching protocol and supervision budget, and identical menus at identical public states; a trainable controller is not reserved for our method.

External comparisons include PropUQ-MAS for inherited-risk scoring, Interlat and LatentMAS for latent communication, and AUQ for explanation-guided control [2, 7, 12, 13]. We will report native behavior separately from matched-controller adaptations. A named paper's missing capability is not silently invented and attributed to its original implementation. CAGE-Cal is a final-output calibration comparator in a compatible shared-answer verification panel, not an unmodified intermediate-task baseline [5]. All extra no-communication runs required by that method are included in

cost. An intervention-trained scalar with receiver/task features and a rich-text intervention head are stronger controls than failure-threshold routing alone [11].

5.3 Three Complementary Evaluations

Fixed-prefix propagation and calibration. All estimators receive the same public prefixes, inherited messages, frozen receiver commitment, and realized reference continuation. The target labels are therefore identical. We will measure Brier score, NLL, error-positive AUROC, AURC, and reliability diagrams for Equation (1), both pooled and by receiver role, edge type, graph depth, and shared-source status. This isolates uncertainty estimation from answer changes. Intermediate local correctness is evaluated only where native labels define it; unavailable local labels are not manufactured by another agent.

Same-state clarification selection. For a preregistered, outcome-independent subset of test prefixes, we will execute the same question menu and continue action under paired seeds. To isolate representation from post-answer execution, diagnostic value heads are fitted for one common frozen rich-text receiver after each reply; these are separate from the policy-specific heads used end to end. We will compare chosen-action utility, regret to the best *measured menu action*, premature stopping when a positive-gain action exists, and unnecessary questions when none does. This finite-menu comparator is not an optimal information-gathering oracle. Candidate-pair prediction, question addressability, unresolved responses, and source-check outcomes are reported separately. Legal receiver tasks that consume the same upstream subproblem provide task-sensitivity pairs; when such pairs are not naturally available, controlled examples are labeled as diagnostics, not native benchmark evidence.

Closed-loop task performance. We will measure final exact match (EM), token F1 where the native evaluator supports it, source-support performance on annotated tasks, and calibration of the actual final answer. The principal trade-off is EM versus atomic questions at budgets $0, 1, 2, 4$, not accuracy at unconstrained additional interaction. Each budget uses independently calibrated terminal scores for its frozen policy. Recovery is the fraction of originally wrong reference outcomes corrected by the intervention policy; regression is the fraction of originally correct outcomes made wrong. Their denominators, unresolved outputs, and all failed runs are retained. Native and matched-controller results are separate; different systems' final-output AUROCs do not constitute a fixed-answer comparison.

5.4 Dependency Propagation and Cross-Role Calibration

Inherited errors and repeated evidence. On naturally erroneous upstream outputs, a masked diagnostic will track whether the same identifiable erroneous premise persists into a receiver artifact and the final answer. The validator uses gold evidence only after execution; instances without an identifiable proposition are reported as unscorable for this diagnostic and remain in the main task evaluation. We will stratify by shared ancestry and source overlap, and measure high-confidence errors along chains and after merges. A separate controlled replay adds an identical already-seen packet without changing evidence. It tests implementation-level duplication invariance, not performance under naturally distinct correlated messages.

Cross-role calibration. We will compare uncalibrated logits, one global calibrator, separate role-only calibrators, and the partially pooled action-family/role calibrator. Prediction features and labels remain fixed. We will report per-role calibration curves and support counts, not only a global ECE. An unseen-role-combination test uses the documented global fallback rather than fitting on test labels. Better calibration cannot be inferred from a higher AUROC, and failure risk is never called the causal contribution of an individual sender.

5.5 Ablations, Cost, and Statistical Analysis

The principal factorial ablation crosses candidate-associated latent content with actual receiver-task conditioning. Further controls remove dependency features, replace candidate latents by ordinary semantic embeddings, pool candidates together, or shuffle candidate–latent associations at test time. Shuffling is a mechanism diagnostic, not a weakened main baseline. Role name alone is compared with the full subtask/state; removing calibration preserves all learned content; disabling clarification preserves the propagation module. We also compare direct rereading at matched compute, a rich-text value head, and generous-text as well as matched-communication budgets. These tests distinguish latent-specific content from additional capacity, ordinary semantic information, and extra computation.

Costs include candidate generation, sender responses, latent pooling, receiver scoring, provenance checks, retrieval, failed calls, and final scoring. We will report model input/output tokens, latent bytes and slots, forward passes, wall-clock latency, peak memory, and training/branch-label cost separately. A latent vector is not counted as a free token, and unshared caching is not credited to the method. Comparisons use the same hardware and parallelism, with equal-question and matched-compute trade-offs rather than an unsupported assertion of equal cost.

All tuning is restricted to development questions, with separate selection and calibration partitions. We will report paired question-group bootstrap intervals using 10,000 resamples; seeds and intervention branches are nested within their original question rather than treated as independent observations. The two primary final-EM comparisons, against rich text and generic latent, use Holm correction at familywise 0.05; subgroup analyses remain descriptive when support is small. A fixed result manifest records every run, including failures and protocol amendments. Tables 2–4 will be populated only from that manifest, without changing the method or evaluation to match test results.

6 Scope and Responsible Use

The method requires access to backbone hidden states and separately fitted adapters when model representations change. Its candidate set, question menu, source annotations, and provenance records bound the information it can use; shared-source features cannot certify independence or source truth. Action calibration is conditional on the reference continuation, and terminal calibration

Table 2: Planned main results for Qwen3-8B. Every entry is unmeasured. A separate table with the same structure will be filled for Mistral-7B. Brier uses the actual final answer and a final-output score; questions count atomic information items. No row implies an implemented or reproduced baseline.

	HotpotQA			MuSiQue			DROP		
Method	EM ↑	Brier ↓	Questions ↓	EM ↑	Brier ↓	Questions ↓	EM ↑	Brier ↓	Questions ↓
Text MAS	—	—	—	—	—	—	—	—	—
Calibrated scalar + control	—	—	—	—	—	—	—	—	—
Rich text + matched control	—	—	—	—	—	—	—	—	—
Generic latent + matched control	—	—	—	—	—	—	—	—	—
PropUQ-MAS + matched control	—	—	—	—	—	—	—	—	—
Interlat + matched control	—	—	—	—	—	—	—	—	—
LatentMAS + matched control	—	—	—	—	—	—	—	—	—
AUQ (native control)	—	—	—	—	—	—	—	—	—
RCLUP	—	—	—	—	—	—	—	—	—

Table 3: Dependency and calibration template with separate scoring and continuation endpoints. All entries are unmeasured. EP: erroneous-premise persistence under each frozen reader, without clarification. HCE: errors given at least 0.9 confidence. Brier/NLL score identical fixed-prefix outcomes; reporting-only recalibration leaves EP unchanged. Denominators are reported alongside results.

Variant	EP ↓	HCE ↓	Brier ↓	NLL ↓
No dependency features	—	—	—	—
Full reader, uncalibrated	—	—	—	—
Full reader, global calibration	—	—	—	—
Full reader, role-only calibration	—	—	—	—
Full reader, partially pooled	—	—	—	—

Table 4: Core ablation template. All cells are unmeasured; role labels alone are not actual task conditioning.

Variant	EM ↑	NLL ↓	Gain ↑
Full RCLUP	—	—	—
Rich text, no candidate latent	—	—	—
Latent, role name only	—	—	—
Rich text, role name only	—	—	—
No dependency features	—	—	—
Generic pooled latent	—	—	—
No inherited-content forwarding	—	—	—
Global calibration only	—	—	—
No clarification	—	—	—

on the deployed policy and evaluation distribution. Neither is a per-instance guarantee. Latent messages can carry sensitive source information and do not provide a privacy boundary. The planned experiments use existing benchmark sources and simulated agent exchanges, not human-user benefit or live-Web safety claims. All source, model, and dataset access terms remain applicable. Human authors must verify the method, citations, experimental records, and any final submission claims.

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A Implementation Specification

A.1 Default Configuration and Interfaces

The following values are declared starting configurations, not selected hyperparameters or measured results. For each invocation, collect $S = 8$ stochastic proposals at temperature 0.7 and top-$p = 0.9$, plus one greedy proposal. Only the eight stochastic proposals contribute counts in Equation (2); the greedy candidate is retained even if its count is zero. Keep at most $K = 4$ distinct candidates, use $\alpha = 0.1$, and retain the residual category. Proposals and grouping are cached and shared by matched methods for a fixed public state. They are generated again only after the state or evidence changes.

Retriever artifacts are canonicalized by ordered document identifiers; interpretation artifacts by normalized entity–relation–value tuples and source identifiers; reasoner artifacts by the dataset's public answer normalization; verifier artifacts by verdict and normalized proposed correction. Order is preserved when semantically relevant. Grouping never uses the test answer. Failure to parse a proposal produces an invalid-proposal record rather than a hidden retry; the shared repair policy determines any retry. The same canonicalizer is used for every representation. Exact canonicalization can over-separate paraphrases; this increases measured candidate fragmentation rather than silently invoking a privileged semantic oracle.

The compressor uses the final backbone layer, $m = 4$ slots per retained candidate, width $d = 256$, and independent linear key/value projections. The residual has four tied learned null slots. The dependency reader contains two four-head pre-layer-normalized attention blocks, each with a width-4d GELU feed-forward layer and residual connections. Candidate text, receiver task, and question encodings are mean pooled from their final-layer frozen-backbone states and projected to d. Head MLPs have two hidden layers of widths 256 and 128, GELU activations, and dropout 0.1. Learned projections are backbone-specific; roles share modules within a backbone.

For slots u, v, the dependency feature vector is

$$d_{uv} = [\mathbb{I}[o_u = o_v], \mathbb{I}[o_u \in A_v], \mathbb{I}[o_v \in A_u], \\ \text{Jaccard}(A_u, A_v), \text{Jaccard}(P_u, P_v), m_{uv}].$$

where o is an origin invocation, A an ancestor set, P a cited-source set, and m a missing-provenance indicator. Empty–empty overlaps equal zero. Exact duplication is keyed by origin, version, and packet content hash before slot expansion. Conflicting content with the same origin/version is an integrity error and is not silently averaged. Exact duplicate coalescing is available to all matched baselines. It is not the novel component of RCLUP.

Every question has one target sender and one explicit candidate distinction. The proposer sees no latent content or gold labels, giving all selectors the same available actions. It emits at most six atomic questions, prioritizing distinct candidate pairs in deterministic support order. A missing alternate is represented by a scope question about the retained candidate versus residual alternatives. Invalid questions are removed before outcome labels are available; their replacement never depends on a branch outcome. The main budget is two atomic clarification items per original task; the sweep uses zero, one, two, and four. Ordinary source fetching, verification, and repair remain active in every arm. Budget exhaustion returns the best checked artifact or UNRESOLVED.

A.2 Source-Backed Clarification and State Updating

The response contains an answer, source IDs, quoted spans, an unresolved flag, and an evidence-event ID. A sender may reread its existing sources or make the same authorized retrieval call available in the reference workflow. Returned quotations must match the identified source version and offsets. This check establishes provenance only: an accurately quoted passage can still be irrelevant or false. No synthetic responder is instructed to reveal a hidden gold answer. Direct source access by the receiver is an explicit compute-matched control.

A new response creates an append-only evidence event. The sender reruns its own role-specific task over the updated accessible history, regenerates proposals and compressed candidate latents, and issues a new message version. A dependency index marks exactly the descendant states that used the replaced version as stale. Recomputed descendants retain independent evidence and are not forced to accept the answer to a clarification. Already committed external actions are not rolled back by this benchmark protocol; the evaluated workflows are information tasks. An identical evidence-event ID is a no-op for belief content, though the attempted call remains in cost. An unresolved response marks the question/source-version pair as attempted and supplies no positive support.

The terminal head uses the same receiver feature architecture at the final verifier state, plus the actual answer encoding. It is trained *after* the deployed policy is frozen. It is never updated on the test answer. For intermediate local-error diagnostics, document-selection support recall uses native annotations, and derived-answer correctness uses native answer labels when the evaluated subtask has them. Arbitrary interpreter claims without a reference are not assigned pseudo-gold local labels. Terminal-rollout supervision in Equations (9)–(10) remains available even when these optional local labels do not exist.

B Shared Workflow and Inference Budget

The task planner decomposes the question into at most three retrieval/interpretation branches using only the question and readable public history. Each branch executes Retriever then Interpreter; the Reasoner combines their outputs and the Verifier checks the proposed answer. A single branch gives a chain; two or three branches give a merge. The Verifier may return a source- or computation-specific issue for at most two ordinary repair rounds, available to every method including π_0. MuSiQue's gold decomposition is never supplied to this planner. No role is asked to vote on the same answer in the principal workflow.

The reproducible corpus is the full released candidate-document pool for each question, not a pool selected by gold support labels. Retrieval uses BM25 ($k_1 = 1.2$, $b = 0.75$), returns three passages per query, and permits direct fetch by document ID. The same six-query allowance and source index are available to all methods. DROP uses its complete supplied paragraph and a common calculator instead of pretending to be open-Web search. This is a closed-corpus Web-evidence setting, not a claim about fresh Web coverage.

Initial shared limits are 65,536 generated and 524,288 processed input tokens per workflow, counting all sampled proposals and responses, and 512 generated tokens per call. The standard atomic-clarification budget is two, with 0, 1, 2, 4 as the controlled sweep. A call that cannot fit the remaining budget is not issued; the workflow retains its unresolved output rather than discarding necessary history. All failed attempts, checks, and repair calls are charged. These are reproducible initial settings to be frozen on development data, not executed costs or a reason to restrict a baseline's normal capabilities.

C Training, Splits, and Calibration Order

The pipeline separates parameters that influence decisions from post-hoc terminal reporting. Official training data are grouped by question/composition family and divided into parameter-fitting, selection, action-calibration, and terminal-calibration portions in a 70:10:10:10 ratio using a fixed hash salt. We cap the initial fitting pool at 2,048 root questions per dataset, chosen without model outcomes; smaller eligible pools are used in full and their actual counts reported. Official labeled development data serve as held-out evaluation and are not used for tuning. Source-disjoint analysis applies the additional support-document exclusion described in Section 5.1. This cap is a declared collection bound, not a claim that any trajectories have been collected.

Stage 1: collect and fit the receiver. The immutable π_0 uses the frozen backbone's ordinary stochastic commitments at temperature 0.7 and top-$p = 0.9$, the fixed role prompts, tools, and shared verification/repair policy. Each seed determines a fresh suffix; question replies use the same decoding settings. Tool execution and scoring are deterministic. Candidate branches intervene only on the current proposed artifact and execute that same continuation. After-clarification prefixes are collected by a question policy fixed before outcomes (round-robin over valid menu entries), not by choosing successful questions. At most two receiver prefixes per training root question are selected by hash with role balancing. Every candidate receives $R = 3$ predetermined continuation seeds. Fit the compressor, recursive forwarding gate, dependency reader, and commit head with Equation (9); the gold answer occurs only in the offline outcome validator. The selected model minimizes question-weighted selection NLL, with question-weighted EM used only as a deterministic tie-breaker.

Stage 2: collect and fit action risks. Freeze the selected receiver before defining the ask macro-action. For each admitted question at a saved prefix, obtain an actual source-backed reply, update the packets, select the new receiver artifact, and execute π_0 using the same three seeds. Train only the question head with Equation (10); inherited representations remain frozen. A recorded continuation can be reused only when its model, complete prompt, tools, source snapshot, candidate commitment, and decoding seed match. Logical invocations and newly executed calls are recorded separately. No sampling trajectory is treated as a new independent question. Policy-specific ask labels can differ when receivers choose different artifacts after the same reply; identical-label claims apply only to the common-receiver selector diagnostic. It uses its own training/calibration targets and is not silently substituted for the deployed head.

Stage 3: calibrate actions and freeze control. Fit Equation (7) on the action-calibration partition using both commit and ask outcomes and root-question weighting. Use L_2 penalty 1 as the default; require 50 calibration prefixes for a group-specific deviation and otherwise use the global map. The penalty, minimum support, and the gain cost λ are selected only through the selection partition. The initial grid for λ is $\{0, 0.01, 0.02, 0.05, 0.10\}$ in final-error-probability units per atomic question. Once selected, the decoder, receiver, question head, action calibrator, source snapshot, budgets, and control policy are frozen.

Stage 4: fit final-output uncertainty. Run the frozen deployed policy on fitting questions to train the terminal head. Fit its global monotone logistic calibrator on the held-out final-calibration split. These questions do not choose a policy, checkpoint, cost, or question threshold. Repeat terminal calibration for each separately reported frozen budget/policy configuration. If the policy changes, recollect policy-specific calibration trajectories; do not reuse an earlier score definition silently. The action-risk and final-risk datasets and model versions are separate artifacts.

Use AdamW with initial learning rate 10^{-4}, weight decay 10^{-2}, batch size 64 prefixes, gradient norm clipping at 1, and at most ten epochs with selection-NLL early stopping. Three fitting seeds are fixed before test evaluation. The initial search is limited to learning rate $\{3{\times}10^{-5}, 10^{-4}, 3{\times}10^{-4}\}$, slots $\{2, 4, 8\}$, and latent width $\{128, 256\}$. This is a development search space, not an instruction to run the full Cartesian product; equal search allocations and every attempted configuration must be logged. Cost and labels for unsuccessful candidates remain in the manifest. Backbone version identifiers, hardware, measured training cost, and executed sample counts belong in the eventual run manifest, not invented table cells.

D Evaluation Definitions and Interpretation

For a collection of error predictions $\hat{r}_n$ and realized binary errors e_n, Brier score is $N^{-1}\sum_n(\hat{r}_n - e_n)^2$ and NLL is the mean Bernoulli negative log likelihood. ECE uses ten equal-width bins; plots include bin counts, and roles with sparse support are not assigned a strong calibration conclusion. Probabilities are clipped only for numerical NLL evaluation at 10^{-6}, consistently across methods. Error is the positive class for AUROC. AURC integrates realized error rate after retaining progressively larger fractions in ascending predicted-risk order; lower is better. Accuracy at 70% coverage is a secondary selective metric. Brier/NLL are calibration-sensitive predictive scores, not by themselves complete proofs of conditional calibration.

In fixed-prefix comparisons, all methods score the same commitment and realized π_0 outcome. In closed-loop comparisons, each method has its own final answer and realized error; these results characterize whole systems and cannot isolate uncertainty ranking. Post-intervention confidence must use the actual post-intervention state. Abstentions and failed executions count as failures for the primary final-success endpoint; selective coverage is reported separately rather than deleting them.

For a replayed prefix and menu action u, define measured utility as $\bar{U}_j(u) = 1 - R^{-1} \sum_r e_{jur} - \lambda c(u)$, with $c(\text{commit}) = 0$. Menu regret is $\max_u \bar{U}_j(u) - \bar{U}_j(\hat{u})$. Report both the mean and paired-seed variability: the maximum of noisy empirical action values is an optimistic finite-sample comparator, not known optimal utility. Gain signs near zero are analyzed with their uncertainty rather than treated as definitive question-necessity labels. Unresolved or source-invalid replies are retained when calculating realized utility.

A supported final answer requires both native answer correctness and the dataset's specified evidence condition when such a condition is available. A quoted span is not sufficient by itself. On datasets without a complete final-evidence validator, EM/F1 and available provenance checks are separate outputs; we do not introduce an unverified composite "grounded success" label. DROP's supplied paragraph supports the arithmetic-transfer study but is not presented as a live retrieval task.

An erroneous premise is scored only when a deterministic native-reference rule can label the identical normalized proposition (including entity and scope); otherwise it is unscorable, not assigned an LLM-generated gold label. HCE is the rate of errors with predicted correctness at least 0.9, using a frozen threshold. HCE uses all eligible cases as its denominator; its erroneous-case denominator is also reported. The propagation audit reports denominators at each level: erroneous upstream propositions, receiver artifacts using those propositions, and final answers using them. Attribution is post-hoc association unless a matched prefix intervention changes only the disputed premise. Natural errors and controlled duplicate-message tests have separate tables. Improvement in average confidence among wrong cases without improving Brier/NLL is not called better calibration. Likewise, fewer questions without equal or better task quality is not called more effective clarification.

E Reproducibility and Falsifiable Claims

The released schema should include question identifier, split, graph, node and role identifiers, subtask hash, model and source versions, parent packet IDs, candidates and support counts, representation configuration, question menu, all observed responses, chosen action, reference-policy version, pre-action risk, final-output risk, native labels, and full cost. Test answer fields are written only by the isolated evaluator. Anonymized release must exclude API keys, local account paths, and identifying source-repository links. Source licensing and redistribution constraints apply independently of model access.

Necessary implementation checks include: normalized candidate support including residual mass; identical-packet coalescing; no gold in agent inputs; provenance offset integrity; unresolved replies not becoming support; idempotent duplicate evidence events; group-calibrator monotonicity; correct fallback for sparse roles; unchanged final answer after post-hoc recalibration; and scores matched to the correct answer/policy version. These checks are specified, not claimed to have been executed for an implemented MAS.

The proposed latent-specific claim fails if structured explanations or matched generic latent content provide the same clarification value within uncertainty. Task-conditioning claims weaken if role labels alone explain the gains. Dependency claims fail if benefits arise only from an exact-duplicate cleanup equally available to all controls. Calibration claims weaken if improvement is confined to pooled ECE while role-wise proper scores worsen. Clarification claims fail if extra questions, a stronger responder, or unequal source access explain the improvement. These are interpretations to retain, not reasons to redesign the test set after evaluation.